



Cochrane
Library

Cochrane Database of Systematic Reviews

Immunosuppressants for IgG4-related disease (Protocol)

Cagnotto G, Bäcklund RT, Bengtsson A, Bruschettini M, supported by Cochrane Sweden

Cagnotto G, Bäcklund RT, Bengtsson A, Bruschettini M, supported by Cochrane Sweden.
Immunosuppressants for IgG4-related disease (Protocol).
Cochrane Database of Systematic Reviews 2025, Issue 8. Art. No.: CD016094.
DOI: [10.1002/14651858.CD016094](https://doi.org/10.1002/14651858.CD016094).

www.cochranelibrary.com

Immunosuppressants for IgG4-related disease (Protocol)

Copyright © 2025 The Authors. Cochrane Database of Systematic Reviews published by John Wiley & Sons, Ltd. on behalf of The Cochrane Collaboration.

WILEY

TABLE OF CONTENTS

ABSTRACT	1
BACKGROUND	2
OBJECTIVES	2
METHODS	2
SUPPLEMENTARY MATERIALS	8
ADDITIONAL INFORMATION	8
REFERENCES	10

[Intervention Protocol]

Immunosuppressants for IgG4-related disease

Giovanni Cagnotto^{1a}, Rebecka Teresia Bäcklund¹, Anders Bengtsson², Matteo Bruschetti³, supported by Cochrane Sweden

¹Rheumatology, Clinical Sciences Malmö, Lund University, Malmö, Sweden. ²Rheumatology, Clinical Sciences Lund, Lund University, Lund, Sweden. ³Cochrane Sweden, Department of Research and Education; Paediatrics, Department of Clinical Sciences Lund, Lund University, Skåne University Hospital, Lund, Sweden, Lund University, Skåne University Hospital, Lund, Sweden

^aORCID iD: <https://orcid.org/0000-0002-1682-8158>

Contact: Giovanni Cagnotto, giovanni.cagnotto@med.lu.se.

Editorial group: Cochrane Central Editorial Service.

Publication status and date: New, published in Issue 8, 2025.

Citation: Cagnotto G, Bäcklund RT, Bengtsson A, Bruschetti M, supported by Cochrane Sweden. Immunosuppressants for IgG4-related disease (Protocol). *Cochrane Database of Systematic Reviews* 2025, Issue 8. Art. No.: CD016094. DOI: [10.1002/14651858.CD016094](https://doi.org/10.1002/14651858.CD016094).

Copyright © 2025 The Authors. Cochrane Database of Systematic Reviews published by John Wiley & Sons, Ltd. on behalf of The Cochrane Collaboration. This is an open access article under the terms of the [Creative Commons Attribution-Non-Commercial Licence](https://creativecommons.org/licenses/by-nc/4.0/), which permits use, distribution and reproduction in any medium, provided the original work is properly cited and is not used for commercial purposes.

ABSTRACT

Objectives

This is a protocol for a Cochrane Review (intervention). The objectives are as follows:

To assess the benefits and harms of immunosuppressants with or without corticosteroids compared to corticosteroid monotherapy for the treatment of IgG4-related disease in people 18 years or older.

BACKGROUND

Description of the condition

IgG4-related disease is an immune-mediated fibro-inflammatory condition that usually involves two or more organs, either simultaneously or in a metachronous pattern [1, 2, 3]. Any organ of the body may be affected by the disease [4]; however, the pancreas, biliary tract, salivary and lacrimal glands, orbits, retroperitoneum and aorta, kidneys, lungs, meninges, and thyroid gland are organs that are more frequently involved [5, 6]. Swelling of the involved organs is the typical manifestation, and it is usually associated with elevation of serum IgG4 subclass levels [4]. Diagnosis is usually confirmed by histological examination of the swollen organ, which typically shows a dense lymphoplasmacytic infiltrate with a high number of IgG4-positive plasma cells, significant fibrosis with a so-called 'storiform' pattern, and obliterative phlebitis [2, 7].

Disease progression is slow, but may lead to organ damage and failure due to advanced fibrosis [3, 4]. IgG4-related disease usually responds to corticosteroid treatment [8], but relapses are common during treatment tapering [9, 10]. Immunosuppressants are often combined with corticosteroids to reduce the number of relapses, thereby reducing the extent of organ damage and the well-known adverse effects of prolonged corticosteroid treatment [3, 11]. However, convincing evidence for this combination treatment strategy is lacking [3], as are extensive epidemiological studies on IgG4-related disease. A Japanese survey estimated an incidence of 0.28 to 1.08 per 100,000 person-years and an estimated prevalence of 6.2 per 100,000 subjects in Japan [12]. A more recent American retrospective cohort study showed an incidence of 1.39 per 100,000 person-years and a prevalence of 5.3 per 100,000 subjects in the USA [13]. IgG4-related disease has a male predominance and typically affects middle-aged and older individuals, with an average age at diagnosis ranging from 50 to 70 years [8, 14, 15, 16]. The disease is generally more severe in males than females [17].

Description of the intervention and how it might work

The typical lesions of IgG4-related disease are characterised by prominent inflammatory infiltrates [4]; therefore, the disease responds well to the anti-inflammatory effects of corticosteroids. Although the pathophysiology of this condition is still not completely understood, experimental data suggest that activation of both innate and adaptive immune responses is a hallmark of IgG-related disease, likely contributing to the fibro-inflammatory process [5, 18, 19, 20, 21, 22]. Treatment with immunosuppressants targeting different components of the immune system may therefore be beneficial. In particular, biologic agents may act through various mechanisms, including B cell depletion, such as with rituximab (anti-CD20) and inebilizumab (anti-CD19), cytokine inhibition, such as with tocilizumab (anti-IL-6), or interference with T cell co-stimulation, as with abatacept, a CTLA-4-Ig fusion protein that modulates the CD80/CD86-CD28 pathway. Conventional synthetic disease-modifying anti-rheumatic drugs (csDMARDs), such as methotrexate and mycophenolate mofetil, exert immunomodulatory effects by interfering with lymphocyte function and inflammatory cytokine production. In addition, small molecule inhibitors such as baricitinib, a Janus kinase (JAK) inhibitor, and rilzabrutinib, a Bruton's tyrosine kinase (BTK) inhibitor, act by blocking intracellular signalling pathways involved in immune cell activation and cytokine production. These mechanisms may lead to disease remission, a reduction in the

frequency of disease flares, a lower cumulative corticosteroid dose, and an overall improvement in the quality of life (QoL) for people with IgG4-related disease. However, immunosuppressive treatments may also carry risks, such as increased susceptibility to serious infection, especially in the elderly, as well as an increased risk of malignancies [23, 24, 25, 26].

Why it is important to do this review

IgG4-related disease was recognised as a distinct disease only 20 years ago [27]. Since then, investigators have been trying to describe the wide spectrum of this clinical entity, as well as working to understand the pathophysiology of this disease. The recent development of comprehensive classification criteria has led to an increased interest in the disease [6]. Clinical trials have recently been conducted, and new promising drugs are in the pipeline. However, due to the complexity of the disease and its recent identification, little is known about the best treatment strategies to adopt, and no approved medications for the treatment of this disease are currently available. IgG4-related disease typically responds well to corticosteroids; however, disease flares are common during corticosteroid tapering, and the optimal tapering regimen remains uncertain. Moreover, prolonged treatment with corticosteroids might lead to untoward adverse effects [28]. Therefore, clinicians often use different immunosuppressants in combination with corticosteroids; however, it is unclear whether these treatment strategies are effective and safe [3, 11]. A previous systematic review using network meta-analysis investigated the use of immunosuppressants in IgG4-related disease. The authors performed a literature search up to February 2019 and only included observational studies; thus, the generated evidence was of low quality [29]. Hence, current treatment recommendations are primarily based on observational studies and expert opinion [3], highlighting the need for high-quality evidence on the management of IgG4-related disease to address this knowledge gap. Given the recent increased awareness of the disease, the recent publication of classification criteria [6], and the surge in the conduct of randomised controlled studies (RCTs), we believe it is now appropriate to undertake this Cochrane systematic review, aiming to generate high-quality evidence that can be useful for clinicians and people dealing with this disease.

OBJECTIVES

To assess the benefits and harms of immunosuppressants with or without corticosteroids compared to corticosteroid monotherapy for the treatment of IgG4-related disease in people 18 years or older.

METHODS

Criteria for considering studies for this review

We will follow the methodological guidance from the *Cochrane Handbook for Systematic Reviews of Interventions* [30] and Methodological Expectations for Cochrane Intervention Reviews [31]. We will report the review following PRISMA guidelines [30, 32, 33].

Types of studies

We will include the following studies.

- RCTs, quasi-RCTs (i.e. parallel-arm RCTs in which allocation is decided by an approximation of randomisation, e.g. allocation by participant ID number), and cross-over RCTs.
- Studies reported as full text, those published only as abstracts, and unpublished data.

We will not include cluster RCTs.

We will not use reporting of our chosen outcomes as a criterion for including or excluding studies.

We will not impose any language restrictions.

Types of participants

We will include participants aged 18 years or older with a diagnosis of IgG4-related disease (also called: IgG4-associated multifocal systemic fibrosis, IgG4-related systemic disease, IgG4-related sclerosing disease, hyper-IgG4 disease, IgG4-related disease, systemic IgG4 plasmacytic syndrome, IgG4-related multi-organ lymphoproliferative syndrome, IgG4-associated disease) according to the 2011 comprehensive diagnostic criteria [34], the American College of Rheumatology/European League Against Rheumatism (ACR/EULAR) 2019 classification criteria [6], or the 2020 revised comprehensive diagnostic criteria [35].

Participants with single organ disease that do not satisfy the above criteria will be included if they fulfil any of the following organ-specific diagnostic criteria:

- IgG4-related autoimmune pancreatitis [36]
- IgG4-related lacrimal gland and saliva adenitis [37]
- IgG4-related kidney disease [38]
- IgG4-related sclerosing cholangitis [39]
- IgG4-related ophthalmic disease [40]
- IgG4-related respiratory disease [41]
- IgG4-related large periarteritis/periarteritis and retroperitoneal fibrosis [42].

Studies including participants younger than 18 will be included, provided the proportion of participants aged 18 years or older is equal to or more than 80%.

We will not apply any restrictions to study settings (e.g. outpatients, hospitalised patients).

Both participants with active disease and those with disease in remission will be included.

Types of interventions

We will include studies investigating the following treatments (either as monotherapy or in combination with corticosteroids).

- Biologics: including but not limited to abatacept, rituximab, tocilizumab, tumour necrosis factor inhibitors
- csDMARDs: including but not limited to azathioprine, iguratimod, leflunomide, methotrexate, mycophenolate mofetil, cyclophosphamide
- JAK inhibitors: including but not limited to baricitinib, filgotinib, tofacitinib
- BTK inhibitors: including but not limited to rilzabrutinib, zanubrutinib.

The comparator of interest is corticosteroid monotherapy.

We will pool different doses and routes of administration of the same interventions (e.g. we will pool rituximab 500 mg and rituximab 1000 mg; intravenous tocilizumab and subcutaneous tocilizumab).

We will not impose any limitations on the dose or route of administration of the interventions and comparisons.

Outcome measures

IgG4-related disease is a recently identified disease; therefore, no recommended outcomes for clinical trials have been established. We will adopt outcomes deemed clinically important by the medical community with experience in IgG4-related diseases, and outcomes that are used in other rheumatic diseases which are similar to IgG4-related disease [3, 43, 44, 45]. When feasible, we will use outcome measures that are based on the validated IgG4-related disease responder index [46, 47].

Critical outcomes

1. Disease flares
2. Disease remission
3. Health-related QoL (HRQoL)
4. Withdrawal due to adverse events
5. Serious adverse events.

Definition of outcomes and endpoints

1. Disease flares

Disease flares are defined as an exacerbation of the disease after an initial improvement. It will be measured as time to first flare, as defined by the trialists.

2. Disease remission

Disease remission is defined as a state of absence of active disease. It will be measured as the proportion of participants with inactive disease, as defined by the trialists.

3. HRQoL

HRQoL is defined as a "multidomain concept that represents the patient's general perception of the effect of illness and treatment on physical, psychological, and social aspects of life" [48]. For the purpose of this review, HRQoL will be measured by the 36-Item Short Form Health Survey Mental Component Summary Score (SF-36 MCS) [49] or by the EQ-5D-5L [50].

4. Withdrawal due to adverse events

An adverse event is defined as any untoward medical occurrence in a clinical investigation subject administered a pharmaceutical product, and which does not necessarily have to have a causal relationship with this treatment [51]. Withdrawal due to adverse events will be measured as the proportion of participants who withdraw from the study treatment due to an adverse event, according to the trialists.

5. Serious adverse events

A serious adverse event is defined as any untoward medical occurrence that meets any of the following criteria [51].

Immunosuppressants for IgG4-related disease (Protocol)

- Results in death
- Life-threatening
- Requires inpatient hospitalisation or prolongation of existing hospitalisation
- Results in persistent or significant disability or incapacity
- Congenital anomaly or birth defect in the offspring of a study participant.

Serious adverse events will be measured as the proportion of participants experiencing at least one serious adverse event, according to the trialists.

Time points for outcome assessment

The chosen outcomes will be evaluated at the latest available time point of each trial.

Important outcomes

These outcomes are important to users of the review.

1. Disease response
2. Cumulative steroid dose.

Definition of outcomes and endpoints

1. Disease response

Disease response is defined as an objective improvement of the disease. It will be measured as the proportion of participants with improvement in the disease, as defined by the trialists and preferably by the IgG4-related disease responder index. Disease response will be evaluated at the latest available time point for each trial.

2. Cumulative steroid dose

Cumulative steroid dose will be reported as the mean dose of prednisolone equivalents. This outcome will be evaluated at the latest available time point for each trial.

Search methods for identification of studies

Electronic searches

We will search the Cochrane Central Register of Controlled Trials (www.cochranelibrary.com), MEDLINE (Pubmed), Embase (Embase.com).

We will search the following trial registers: the U.S. National Institutes of Health clinical trials registry (clinicaltrials.gov), the EU Clinical Trials Register (eudract.ema.europa.eu), and the World Health Organization (WHO) trials portal (trialsearch.who.int).

For assessment of adverse events, we will also search for the websites of the regulatory agencies U.S. Food and Drug Administration-MedWatch (www.fda.gov/safety/medwatch-fda-safety-information-and-adverse-event-reporting-program), European Medicines Evaluation Agency (www.ema.europa.eu), Australian Adverse Drug Reactions Bulletin (www.tga.gov.au/safety), and UK Medicines and Healthcare products Regulatory Agency for pharmacovigilance and drug safety updates (www.mhra.gov.uk).

We will search all databases from their inception to the present, and we will pose no language restrictions.

The search strategy is detailed in [Supplementary material 1](#).

Searching other resources

We will check reference lists of all primary studies and review articles for additional references. We will search relevant manufacturer websites for trial information. We will search for errata or retractions from included studies published in full text on PubMed (www.ncbi.nlm.nih.gov/pubmed), and report the date this was done within the review.

Data collection and analysis

Selection of studies

Two review authors (GC, RTB) will independently screen titles and abstracts for inclusion of all the potentially relevant studies we identify as a result of the search, and code them as 'retrieve' (eligible or potentially eligible/unclear) or 'do not retrieve'. We will retrieve the full-text study report publications, and two review authors (GC, RTB) will independently screen the full-text and identify studies for inclusion, and identify and record reasons for exclusion of the ineligible studies. We will resolve any disagreement through discussion or, if required, we will consult a third review author (MB or AB). If necessary, we will contact study authors by email to obtain missing information relevant to determining study eligibility. We will identify and exclude duplicates and collate multiple reports of the same study so that each study, rather than each report, is the unit of interest in the review. We will record the selection process in sufficient detail to complete a PRISMA flow diagram [32], and the 'Characteristics of excluded studies' table.

We will manage the results of the search using Covidence [52].

Data extraction and management

We will use a data collection form for study characteristics and outcome data, which we will pilot-test on at least one included study in the review. One review author (GC) will extract study characteristics from the included studies. A second review author (RTB) will spot-check study characteristics for accuracy against the trial report. We will extract the following study characteristics.

- Methods: study design, total duration of study, details of any 'run-in' period, number of study centres and location, study setting, withdrawals, and date of study.
- Participants: number, mean age, age range, sex, disease duration, severity of condition, diagnostic criteria, inclusion criteria, and exclusion criteria.
- Interventions: intervention (including dose, route of administration and duration), comparison (including dose, route of administration and duration), concomitant medications, and excluded medications.
- Outcomes: critical and important outcomes specified and collected, and time points reported.
- Characteristics of the design of the study as outlined below in the [Risk of bias assessment in included studies](#) section
- Notes: study funding and notable declarations of interest of trial authors.

Two review authors (GC, RTB) will independently extract outcome data from the included studies. We will extract the number of events and number of participants per treatment group for dichotomous

outcomes, and the means, standard deviations and number of participants per treatment group for continuous outcomes. We will extract the number of participants, hazard ratios (HRs) and standard errors for time-to-event data (e.g. disease flares). If necessary, we will extract HRs from Kaplan-Meier curves or from other statistics in each study, using standard methods [53]. We will note in the 'Characteristics of included studies' table if outcome data were not reported in a usable way, and when data were transformed or estimated from a graph. We will resolve disagreements by consensus or by involving a third review author (AB). One review author (GC) will transfer data into Review Manager [54]. We will double-check that the data are entered correctly by comparing the data presented in the systematic review with the study reports. Should any queries arise, or in cases where additional data are required, we will contact study investigators by email for clarification. We will use PlotDigitizer (PlotDigitizer — Extract Data from Graph Image Online) to extract data from graphs or figures. These data will also be extracted in duplicate. We will analyse all participants randomised on an intention-to-treat (ITT) basis.

Where reports include multiple measures of a single outcome, our order of preference will be as follows.

Disease flares

- The definition of flare reported by the trialists and based on the IgG4-related disease responder index [46, 47].
- The definition of flare reported by the trialists that correlates best with the definition reported in other included studies.

Disease remission

- The definition of remission reported by the trialists and based on the IgG4-related disease responder index.
- The most stringent definition of remission provided by the trialists and that correlates best with the definition reported in other included studies.

HRQoL

- SF-36 MCS
- EQ-5D-5L.

Disease response

- The definition of disease response reported by the trialists and based on the IgG4-related disease responder index.
- The less stringent definition of disease response reported by the trialists and that correlates best with the definition reported in other included studies.

Risk of bias assessment in included studies

Two review authors (GC, RTB) will independently assess the risk of bias (RoB) in each included study for each specific outcome to which the same study contributes, using the RoB 2 tool [55] and the criteria outlined in the *Cochrane Handbook for Systematic Reviews of Intervention* [56]. Any disagreements will be resolved through discussion or by involving another review author (MB).

We will investigate the effect of assignment to the intervention.

The following domains will be considered.

- Bias arising from the randomisation process
- Bias due to deviations from the intended interventions (effect of assignment to intervention)
- Bias due to missing outcome data
- Bias in the measurement of the outcome
- Bias in the selection of the reported result.

RoB assessment will be applied to all the critical outcomes. Each potential source of bias will be classified as 'high risk', 'low risk', or 'some concerns', and a justification will be provided for each judgment in the RoB table. A summary of our judgments across studies for each of the five domains listed will be provided.

Overall RoB judgment will be provided for each study, according to the following rules [56].

- Low risk: the study is judged to be at low risk for all domains for the considered outcome
- Some concerns: the study is judged to raise some concerns in at least one domain for the considered outcome, but is not at high risk for any domain
- High risk: the study is judged to be at high risk in at least one domain for the considered outcome or when the study is judged to have some concerns for multiple domains in a way that substantially reduces confidence in the outcome.

We will contact study authors by email to obtain clarification of potential inconsistencies in relevant information for RoB.

Where information on RoB is related to unpublished data or is obtained by correspondence with a trialist, we will note this in the 'Risk of bias' table.

A RoB graph and RoB summary will be presented for each outcome considered in the review.

Assessment of bias in conducting the systematic review

We will conduct the review according to this published protocol. Any deviations will be reported in the 'Differences between protocol and review' section of the systematic review.

Measures of treatment effect

We will analyse dichotomous data as risk ratios (RRs) or Peto odds ratios (ORs) when the outcome is a rare event (approximately < 10%), and use 95% confidence intervals (CIs). Time-to-event data (e.g. disease flares) will be analysed as HRs and 95% CIs, whereas continuous data will be analysed by the mean difference (MD) and 95% CIs. We will enter data presented as a scale with a consistent direction of effect across studies.

When different scales are used to measure the same conceptual outcome (e.g. HRQoL), standardised mean differences (SMDs) will be calculated, with corresponding 95% CIs. SMDs will be back-translated to a typical scale (e.g. 0 to 100 for HRQoL) by multiplying the SMD by a typical among-person standard deviation (e.g. the standard deviation of the control group at baseline from the most representative trial) as per the *Cochrane Handbook for Systematic Reviews of Interventions* [57].

In the 'Effect of interventions' section and the 'Comments' column of the summary of findings table, we will provide the absolute per cent difference and the relative per cent change from baseline risk

Immunosuppressants for IgG4-related disease (Protocol)

Copyright © 2025 The Authors. Cochrane Database of Systematic Reviews published by John Wiley & Sons, Ltd. on behalf of The Cochrane Collaboration.

for dichotomous outcomes. We will calculate the absolute per cent change from the difference in the risks between the intervention and control group using GRADEpro GDT [58] and expressed as a percentage. The relative per cent change for dichotomous data will be calculated as the RR - 1 and expressed as a percentage.

In the 'Effect of interventions' section and the 'Comments' column of the summary of findings table, we will provide the absolute risk of flare-free survival for time-to-event data. We will calculate the intervention risk using the method by Tierney and colleagues [59].

In the 'Effect of interventions' section and the 'Comments' column of the summary of findings tables, we will provide the absolute benefit of continuous outcomes, which will be calculated as the improvement in the intervention group minus the improvement in the control group, expressed in the original units.

When interpreting HRQoL, we will use +2.4 and -1.2 points of SF-36 MCS as minimal clinically important difference (MCID) for improvement and worsening, respectively. These cut-offs were calculated in a cohort of individuals with systemic lupus erythematosus (SLE) by Colangelo and colleagues [60]. SLE is a chronic autoimmune disease, and, like IgG4-related disease, it usually involves multiple organs. Given the similarities between these two diseases, and due to the lack of an established MCID of SF-36 for IgG4-related disease, we find it appropriate to use the cut-offs reported above in our review.

Unit of analysis issues

Where multiple trial arms are reported in a single trial, we will include only the relevant arms. If two comparisons are combined in the same meta-analysis, we will halve the control group to avoid double-counting. In the event that we identify cross-over trials, in which the reporting of continuous outcome data precludes paired analysis, we will not include these data in a meta-analysis, in order to avoid unit of analysis error. Where carry-over effects are thought to exist, and where sufficient data exists, we will only include data from the first period in the analysis.

Dealing with missing data

Where possible, we will contact investigators or study sponsors by email to verify key study characteristics and obtain missing numerical outcome data (e.g. when a study is identified as abstract only, or when data is not available for all participants). Where this is not possible, and the missing data are thought to introduce serious bias, we will explore the impact of including such studies in the overall assessment of results by a sensitivity analysis. Any assumptions and imputations used to handle missing data will be clearly described, and the effect of imputation will be explored by sensitivity analyses. For dichotomous outcomes (e.g. disease remission), the denominator will be the number of participants in each group for which the outcome was measured. If this number is not available, we will use the number of participants randomised in each intervention group as the denominator. For continuous outcomes (e.g. HRQoL), we will calculate the MD or SMD based on the number of participants analysed at that time point. If the number of participants analysed is not presented for each time point, the number of randomised participants in each group at baseline will be used. For time-to-event data, HR will be calculated based on the number of randomised participants in each treatment arm. Where possible, missing standard deviations will be computed from other statistics such as standard errors,

CI or P values, according to the methods recommended in the *Cochrane Handbook for Systematic Reviews of Interventions* [30]. If standard deviations cannot be calculated, they will be imputed (e.g. from other studies in the meta-analysis). For outcomes assessing adverse events, we will extract data for those randomised to and receiving the allocated treatment or comparison.

Reporting bias assessment

To assess outcome reporting bias, two authors (GC, RTB) will check trial protocols against published reports. We will search the WHO International Clinical Trials Registry Platform (<https://trialsearch.who.int>), and the U.S. National Institutes of Health clinical trials registry (<https://clinicaltrials.gov>) for the a priori trial protocol. If trial protocols are unavailable, we will compare the outcomes reported in the methods and results sections of the trial reports. We will generate funnel plots for meta-analyses, including at least 10 trials of varying sizes to assess the effects of small studies. If we detect asymmetry in the funnel plot, we will review the characteristics of the trials to assess whether the asymmetry is likely due to publication bias or other factors, such as the methodological or clinical diversity of the trials. Where we are able to pool more than 10 trials, we will conduct formal statistical tests to investigate funnel plot asymmetry and follow the recommendations in Chapter 13 of the *Cochrane Handbook for Systematic Reviews of Interventions* [61].

Synthesis methods

We will undertake meta-analyses only where this is meaningful, that is, if the treatments, participants and the underlying clinical question are similar enough for pooling to make sense. If a meta-analysis is not possible, we will undertake a narrative description of the studies included in the review, following the Synthesis Without Meta-analysis (SWiM) reporting guidelines [62].

We will pool different doses and routes of administration of the same drug (e.g. rituximab 1000 mg and rituximab 500 mg; intravenous tocilizumab and subcutaneous tocilizumab).

We will use a random-effects model in the primary analysis, since we expect differences in treatment effect across studies, due to the heterogeneity of the disease and the heterogeneity of drugs within each intervention group. We will also perform a sensitivity analysis using a fixed-effect model.

The primary analysis will include all the studies, with outcomes assessed at the latest available time point for each study. This approach should maximise statistical power.

The planned comparisons are (in order of importance):

1. Biologic monotherapy or combined with corticosteroids versus corticosteroid monotherapy
2. csDMARD monotherapy or combined with corticosteroids versus corticosteroid monotherapy
3. JAK-inhibitor monotherapy or combined with corticosteroids versus corticosteroid monotherapy
4. BTK-inhibitor monotherapy or combined with corticosteroids versus corticosteroid monotherapy.

Immunosuppressants for IgG4-related disease (Protocol)

Investigation of heterogeneity and subgroup analysis

Clinical and methodological diversity will be assessed in terms of participants, interventions, outcomes and study characteristics for the included studies to determine whether a meta-analysis is appropriate. This will be conducted by observing the data from the data extraction tables. Statistical heterogeneity will be assessed by visual inspection of the forest plot to assess for obvious differences in results between the studies, and using the I^2 and χ^2 statistical tests. As recommended in the *Cochrane Handbook for Systematic Reviews of Interventions* [30], the interpretation of an I^2 value of 0% to 40% might 'not be important'; 30% to 60% may represent 'moderate' heterogeneity; 50% to 90% may represent 'substantial' heterogeneity; and 75% to 100% represents 'considerable' heterogeneity. As noted in the *Cochrane Handbook for Systematic Reviews of Interventions*, we will keep in mind that the importance of I^2 depends on: (i) magnitude and direction of effects and (ii) strength of evidence for heterogeneity. The χ^2 test will be interpreted where a P value < 0.10 will indicate evidence of statistical heterogeneity. If we identify substantial heterogeneity, we will report it and investigate possible causes by following the recommendations in Chapter 10, Section 10 of the *Cochrane Handbook for Systematic Reviews of Interventions* [63].

Since the mechanisms of action of the different drugs within the main intervention groups (biologics and csDMARDs) may vary significantly (e.g. abatacept versus rituximab; methotrexate versus mycophenolate mofetil), we will perform subgroup analyses to investigate the effects of different mechanisms of action, or individual drugs, within these groups, compared to corticosteroid monotherapy. Subgroups may include, for example, B-cell depleting therapies, IL-6 inhibitors, purine and pyrimidine synthesis inhibitors, folate metabolism inhibitors, and alkylating agents.

Including trials investigating single-organ disease and multi-organ disease in the same analysis could introduce heterogeneity due to potential differences in prognosis and treatment response. Therefore, we will consider performing a subgroup analysis, stratifying studies with more than 50% of participants with single-organ disease versus studies with more than 50% of participants with multi-organ disease, should the primary analysis reveal significant heterogeneity.

The outcomes of choice for subgroup analyses will be disease flares and withdrawals due to adverse events.

We will use the formal test for subgroup interactions in Review Manager [54] and will use caution in the interpretation of subgroup analysis as advised in Chapter 10 of the *Cochrane Handbook for Systematic Reviews of Interventions* [63]. The magnitude of the effects will be compared between the subgroups by means of assessing the overlap of the CIs with the summary estimate. Non-overlap of the CIs indicates statistical significance.

Equity-related assessment

IgG4-related disease is a relatively newly recognised condition, and most current research efforts are focused on understanding its clinical features and evaluating core treatment strategies. As such, equity-related variables (such as ethnicity, socioeconomic status, religion, or sexual orientation) are typically not collected or reported in published studies. Therefore, we believe it is

premature to systematically assess equity issues in this context. Furthermore, a meaningful equity-focused assessment would require an individual participant data systematic review, which is beyond the scope of our work. Nevertheless, we acknowledge that IgG4-related disease predominantly affects older adults, a population that may be considered vulnerable and, where possible, we will consider age-related implications in our interpretation of the findings.

Sensitivity analysis

To explore the robustness of our findings, we will perform a sensitivity analysis for the outcome disease flares by excluding studies at overall high RoB. We will also perform a sensitivity analysis using a fixed-effect model, including all the studies.

Sensitivity analyses reported above will only be performed for the comparison of biologic monotherapy or combined with corticosteroids versus corticosteroid monotherapy.

The primary analysis will include outcomes that are assessed at the latest available time point in each study. This method aims to maximise statistical power. However, it may generate biased estimates for time-dependent outcomes (e.g. disease remission), especially if the time points are very different between the studies. Therefore, we will perform a sensitivity analysis stratified by study duration (≤ 6 months versus > 6 months), but only if the primary analysis reveals substantial heterogeneity for disease remission that cannot be attributed to other factors (e.g. differences in treatment). The outcome disease flares is a time-to-event outcome, and it should not be affected, provided the assumption of proportional hazard is fulfilled. Therefore, we will assess the plausibility of the proportional hazards assumption by examining the Kaplan–Meier curves reported in the included studies, where available. If the curves show approximately proportional separation over time, we will consider the assumption reasonable. If substantial violations are suspected, we will conduct a sensitivity analysis excluding those studies.

These sensitivity analyses will be performed for all the comparisons.

Certainty of the evidence assessment

We will create summary of findings tables using the following outcomes.

- Disease flares
- Disease response
- HRQoL
- Withdrawal due to adverse events
- Serious adverse events.

Time points of interest will be within a medium- to long-term time frame (e.g. up to 12 months). Details are specified in the section [Outcome measures](#) and [Synthesis methods](#).

Key comparisons of interest will be:

1. Biologics versus corticosteroid monotherapy
2. csDMARDs versus corticosteroid monotherapy.

Two review authors (GC, MB) will independently assess the certainty of the evidence. We will resolve disagreements by

Immunosuppressants for IgG4-related disease (Protocol)

consensus or by involving a third review author (AB). We will use the five GRADE (Grades of Recommendation, Assessment, Development and Evaluation) considerations (study limitations, consistency of effect, imprecision, indirectness and publication bias) to assess the certainty of a body of evidence as it relates to the studies which contribute data to the meta-analyses for the prespecified outcomes, and report the certainty of evidence as high, moderate, low, or very low [64].

- High: we are very confident that the true effect lies close to that of the estimate of the effect.
- Moderate: we are moderately confident in the effect estimate; the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
- Low: our confidence in the effect estimate is limited; the true effect may be substantially different from the estimate of the effect.
- Very low: we have very little confidence in the effect estimate; the true effect is likely to be substantially different from the estimate of effect.

We will use the methods and recommendations described in Chapter 14 of the *Cochrane Handbook for Systematic Reviews of Interventions* [65]. We will use GRADEpro software to prepare the summary of findings tables [58]. We will justify all decisions to downgrade the certainty of studies using footnotes, and we will make comments to aid the reader's understanding of the review where necessary. We will provide absolute and relative percentage change in the 'Comments' column of the summary of findings tables, as described in [Measures of treatment effect](#).

Consumer involvement

When developing this protocol, we collaborated with Mr Lars Boustedt, who is a person living with IgG4-related disease, and who has been followed up at Skåne University Hospital in Lund, Sweden, since 2012. We will contact him again during review development for further feedback on the review findings and conclusions.

SUPPLEMENTARY MATERIALS

Supplementary materials are available with the online version of this article: [10.1002/14651858.CD016094](https://doi.org/10.1002/14651858.CD016094).

Supplementary material 1 Search strategies

ADDITIONAL INFORMATION

Acknowledgements

We acknowledge Matthias Bank and Maria Björklund, Library and ICT services, Lund University, Sweden, for defining the search strategies (see [Supplementary material 1](#)). We thank Mr Lars Boustedt (Lund, Sweden) for his comments on the protocol as a patient living with IgG4-related disease.

Editorial and peer-reviewer contributions

Cochrane Neonatal supported the authors in the development of this protocol.

The following people conducted the editorial process for this protocol:

- Sign-off Editor (final editorial decision): Toby Lasserson, Acting Editor in Chief of The Cochrane Library;
- Managing Editor (selected peer reviewers, provided editorial guidance to authors, edited the article): Alan Thomas, Cochrane Central Editorial Service;
- Editorial Assistant (conducted editorial policy checks, collated peer-reviewer comments and supported editorial team): Lisa Wydrzynski, Cochrane Central Editorial Service;
- Copy Editor (copy editing and production): Narelle Willis, Cochrane Central Production Service;
- Peer-reviewers (provided comments and recommended an editorial decision): Ichiro Mizushima, Department of Nephrology and Rheumatology, Kanazawa University Hospital (clinical/content review), Guy Katz, Massachusetts General Hospital, Harvard Medical School (clinical/content review), Brian Duncan (consumer review), Clare Miles, Cochrane Evidence Production and Methods Directorate (methods review), Jo Platt, Cochrane Evidence Production and Methods Directorate (search review). One additional peer reviewer provided clinical peer review but chose not to be publicly acknowledged.

Contributions of authors

GC conceived, designed and wrote the protocol. RB, AB and MB critically reviewed and discussed the protocol with GC. All authors approved the final version to be published.

Declarations of interest

GC: AlfaSigma S.p.A. (Grant/Contract); payment made to home institution with no financial benefits from this payment and did not have access to or control of the funds; works as Senior Consultant Rheumatologist, Skåne University Hospital Malmö, Sweden.

MB: no relevant interests; Senior Editor of Cochrane Neonatal and was not involved in the editorial process or decision-making of this protocol.

RB: no relevant interests; works as a general physician (AT-läkare) in Helsingborg.

AB: none known.

Sources of support

Internal sources

- Institute for Clinical Sciences, Lund University, Lund, Sweden
- MB is employed by this organization.

External sources

- Region Skåne, Skåne University Hospital and Lund University, Sweden

Cochrane Sweden is supported from Region Skåne, Skåne University Hospital and Lund University

Registration and protocol

Cochrane approved the proposal for this review in December 2023.

Data, code and other materials

Data sharing is not applicable to this article as it is a protocol, so no datasets were generated or analysed.

REFERENCES

1. Umehara H, Okazaki K, Masaki Y, Kawano M, Yamamoto M, Saeki T, et al. A novel clinical entity, IgG4-related disease (IgG4RD): general concept and details. *Modern Rheumatology* 2012;**22**(1):1-14. [DOI: [10.1007/s10165-011-0508-6](https://doi.org/10.1007/s10165-011-0508-6)]
2. Kamisawa T, Zen Y, Pillai S, Stone JH. IgG4-related disease. *Lancet* 2015;**385**(9976):1460-71. [DOI: [10.1016/S0140-6736\(14\)60720-0](https://doi.org/10.1016/S0140-6736(14)60720-0)]
3. Khosroshahi A, Wallace ZS, Crowe JL, Akamizu T, Azumi A, Carruthers MN, et al. International consensus guidance statement on the management and treatment of IgG4-related disease. *Arthritis & Rheumatology* 2015;**67**(7):1688-99. [DOI: [10.1002/art.39132](https://doi.org/10.1002/art.39132)]
4. Stone JH, Zen Y, Deshpande V. IgG4-related disease. *New England Journal of Medicine* 2012;**366**(6):539-51. [DOI: [10.1056/NEJMr1104650](https://doi.org/10.1056/NEJMr1104650)]
5. Perugino CA, Stone JH. IgG4-related disease: an update on pathophysiology and implications for clinical care. *Nature Reviews Rheumatology* 2020;**16**(12):702-14. [DOI: [10.1038/s41584-020-0500-7](https://doi.org/10.1038/s41584-020-0500-7)]
6. Wallace ZS, Naden RP, Chari S, Choi H, Della-Torre E, Dicaire JF, et al. The 2019 American College of Rheumatology/ European League Against Rheumatism classification criteria for IgG4-related disease. *Arthritis & Rheumatology* 2020;**72**(1):7-19. [DOI: [10.1002/art.41120](https://doi.org/10.1002/art.41120)]
7. Deshpande V, Zen Y, Chan JK, Yi EE, Sato Y, Yoshino T, et al. Consensus statement on the pathology of IgG4-related disease. *Modern Pathology* 2012;**25**(9):1181-92. [DOI: [10.1038/modpathol.2012.72](https://doi.org/10.1038/modpathol.2012.72)]
8. Inoue D, Yoshida K, Yoneda N, Ozaki K, Matsubara T, Nagai K, et al. IgG4-related disease: dataset of 235 consecutive patients. *Medicine* 2015;**94**(15):e680. [DOI: [10.1097/MD.0000000000000680](https://doi.org/10.1097/MD.0000000000000680)]
9. Kamisawa T, Okazaki K. Diagnosis and treatment of IgG4-related disease. *Current Topics in Microbiology & Immunology* 2017;**401**:19-33. [DOI: [10.1007/82_2016_36](https://doi.org/10.1007/82_2016_36)]
10. Raina A, Yadav D, Krasinskas AM, McGrath KM, Khalid A, Sanders M, et al. Evaluation and management of autoimmune pancreatitis: experience at a large US center. *American Journal of Gastroenterology* 2009;**104**(9):2295-306. [DOI: [10.1038/ajg.2009.325](https://doi.org/10.1038/ajg.2009.325)]
11. Lanzillotta M, Mancuso G, Della-Torre E. Advances in the diagnosis and management of IgG4 related disease. *BMJ* 2020;**369**:m1067. [DOI: [10.1136/bmj.m1067](https://doi.org/10.1136/bmj.m1067)]
12. Uchida K, Masamune A, Shimosegawa T, Okazaki K. Prevalence of IgG4-related disease in Japan based on nationwide survey in 2009. *International Journal of Rheumatology* 2012;**2012**:358371. [DOI: [10.1155/2012/358371](https://doi.org/10.1155/2012/358371)]
13. Wallace ZS, Miles G, Smolkina E, Petruski-Ivleva N, Madziva D, Cook C, et al. Incidence, prevalence and mortality of IgG4-related disease in the USA: a claims-based analysis of commercially insured adults. *Annals of the Rheumatic Diseases* 2023;**82**(7):957-62. [DOI: [10.1136/ard-2023-223950](https://doi.org/10.1136/ard-2023-223950)]
14. Jha I, McMahon GA, Perugino CA, Katz G, Wallace ZS, Fernandes A, et al. Sex as a predictor of clinical phenotype and determinant of immune response in IgG4-related disease: a retrospective study of 328 patients fulfilling the American College of Rheumatology-European League Against Rheumatism classification criteria. *2024 Lancet Rheumatology*;6(7):e460-8. [DOI: [10.1016/S2665-9913\(24\)00089-4](https://doi.org/10.1016/S2665-9913(24)00089-4)]
15. Wallace ZS, Deshpande V, Mattoo H, Mahajan VS, Kulikova M, Pillai S, et al. IgG4-related disease: clinical and laboratory features in one hundred twenty-five patients. *Arthritis & Rheumatology* 2015;**67**(9):2466-75. [DOI: [10.1002/art.39205](https://doi.org/10.1002/art.39205)]
16. Wang L, Zhang P, Zhang X, Lin W, Tang H, Li J, et al. Sex disparities in clinical characteristics and prognosis of immunoglobulin G4-related disease: a prospective study of 403 patients. *Rheumatology* 2019;**58**:820-30. [DOI: [10.1093/rheumatology/key397](https://doi.org/10.1093/rheumatology/key397)]
17. Stone JH. IgG4-related disease: lessons from the first 20 years. *Rheumatology* 2025;**64**(Suppl 1):i24-7. [DOI: [10.1093/rheumatology/keaf008](https://doi.org/10.1093/rheumatology/keaf008)]
18. Mattoo H, Mahajan VS, Della-Torre E, Sekigami Y, Carruthers M, Wallace ZS, et al. De novo oligoclonal expansions of circulating plasmablasts in active and relapsing IgG4-related disease. *Journal of Allergy & Clinical Immunology* 2014;**134**(3):679-87. [DOI: [10.1016/j.jaci.2014.03.034](https://doi.org/10.1016/j.jaci.2014.03.034)]
19. Maehara T, Mattoo H, Ohta M, Mahajan VS, Moriyama M, Yamauchi M, et al. Lesional CD4+ IFN-γ+ cytotoxic T lymphocytes in IgG4-related dacryoadenitis and sialoadenitis. *Annals of the Rheumatic Diseases* 2017;**76**(2):377-85. [DOI: [10.1136/annrheumdis-2016-209139](https://doi.org/10.1136/annrheumdis-2016-209139)]
20. Hubers LM, Vos H, Schuurman AR, Erken R, Oude Elferink RP, Burgering B, et al. Annexin A11 is targeted by IgG4 and IgG1 autoantibodies in IgG4-related disease. *Gut* 2018;**67**(4):728-35. [DOI: [10.1136/gutjnl-2017-314548](https://doi.org/10.1136/gutjnl-2017-314548)]
21. Furukawa S, Moriyama M, Tanaka A, Maehara T, Tsuboi H, Iizuka M, et al. Preferential M2 macrophages contribute to fibrosis in IgG4-related dacryoadenitis and sialoadenitis, so-called Mikulicz's disease. *Clinical Immunology* 2015;**156**(1):9-18. [DOI: [10.1016/j.clim.2014.10.008](https://doi.org/10.1016/j.clim.2014.10.008)]
22. Akiyama M, Suzuki K, Yamaoka K, Yasuoka H, Takeshita M, Kaneko Y, et al. Number of circulating follicular helper 2 T cells correlates with IgG4 and interleukin-4 levels and plasmablast numbers in IgG4-related disease. *Arthritis & Rheumatology* 2015;**67**(9):2476-81. [DOI: [10.1002/art.39209](https://doi.org/10.1002/art.39209)]
23. Bilal J, Berlinberg A, Riaz IB, Faridi W, Bhattacharjee S, Ortega G, et al. Risk of infections and cancer in patients with rheumatologic diseases receiving interleukin inhibitors: a systematic review and meta-analysis [Erratum in: JAMA Netw Open. 2020 Apr 1;3(4):e205507.

doi: 10.1001/jamanetworkopen.2020.5507]. *JAMA Network Open* 2019;**2**(10):e1913102. [DOI: [10.1001/jamanetworkopen.2019.13102](https://doi.org/10.1001/jamanetworkopen.2019.13102)]

24. Bongartz T, Sutton AJ, Sweeting MJ, Buchan I, Matteson EL, Montori V. Anti-TNF antibody therapy in rheumatoid arthritis and the risk of serious infections and malignancies: systematic review and meta-analysis of rare harmful effects in randomized controlled trials. *JAMA* 2006;**295**(19):2275-85. [DOI: [10.1001/jama.295.19.2275](https://doi.org/10.1001/jama.295.19.2275)]

25. Gottenberg JE, Ravaud P, Bardin T, Cacoub P, Cantagrel A, Combe B, et al. Risk factors for severe infections in patients with rheumatoid arthritis treated with rituximab in the autoimmunity and rituximab registry. *Arthritis & Rheumatism* 2010;**62**(9):2625-32. [DOI: [10.1002/art.27555](https://doi.org/10.1002/art.27555)]

26. Ebbo M, Grados A, Samson M, Groh M, Loundou A, Rigolet A, et al. Long-term efficacy and safety of rituximab in IgG4-related disease: data from a French nationwide study of thirty-three patients. *PLoS ONE [Electronic Resource]* 2017;**12**(9):e0183844. [DOI: [10.1371/journal.pone.0183844](https://doi.org/10.1371/journal.pone.0183844)]

27. Kamisawa T, Funata N, Hayashi Y, Eishi Y, Koike M, Tsuruta K, et al. A new clinicopathological entity of IgG4-related autoimmune disease. *Journal of Gastroenterology* 2003;**38**(10):982-4. [DOI: [10.1007/s00535-003-1175-y](https://doi.org/10.1007/s00535-003-1175-y)]

28. Masaki Y, Shimizu H, Sato Nakamura T, Nakamura T, Nakajima A, Iwao Kawanami H, et al. IgG4-related disease: diagnostic methods and therapeutic strategies in Japan. *Journal of Clinical & Experimental Hematopathology* 2014;**54**(2):95-101. [DOI: [10.3960/jslrt.54.95](https://doi.org/10.3960/jslrt.54.95)]

29. Omar D, Chen Y, Cong Y, Dong L. Glucocorticoids and steroid sparing medications monotherapies or in combination for IgG4-RD: a systematic review and network meta-analysis. *Rheumatology* 2020;**59**(4):718-26. [DOI: [10.1093/rheumatology/kez380](https://doi.org/10.1093/rheumatology/kez380)]

30. Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.5 (updated August 2024). Cochrane, 2024. Available from <https://www.training.cochrane.org/handbook>.

31. Higgins JP, Lasserson T, Thomas J, Flemming E, Churchill R. *Methodological expectations of Cochrane intervention reviews*. Cochrane: London, Version August 2023. Available from <https://community.cochrane.org/mecir-manual> 2023.

32. Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ* 2021;**372**:n71. [DOI: [10.1136/bmj.n71](https://doi.org/10.1136/bmj.n71)]

33. Page MJ, Moher D, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. PRISMA 2020 explanation and elaboration: updated guidance and exemplars for reporting systematic reviews. *BMJ* 2021;**372**:n160. [DOI: [10.1136/bmj.n160](https://doi.org/10.1136/bmj.n160)]

34. Umehara H, Okazaki K, Masaki Y, Kawano M, Yamamoto M, Saeki T, et al. Comprehensive diagnostic criteria for IgG4-

related disease (IgG4-RD), 2011. *Modern Rheumatology* 2012;**22**(1):21-30. [DOI: [10.1007/s10165-011-0571-z](https://doi.org/10.1007/s10165-011-0571-z)]

35. Umehara H, Okazaki K, Kawa S, Takahashi H, Goto H, Matsui S, et al. The 2020 revised comprehensive diagnostic (RCD) criteria for IgG4-RD. *Modern Rheumatology* 2021;**31**(3):529-33. [DOI: [10.1080/14397595.2020.1859710](https://doi.org/10.1080/14397595.2020.1859710)]

36. Shimosegawa T, Chari ST, Frulloni L, Kamisawa T, Kawa S, Mino-Kenudson M, et al. International consensus diagnostic criteria for autoimmune pancreatitis: guidelines of the International Association of Pancreatology. *Pancreas* 2011;**40**(3):352-8. [DOI: [10.1097/MPA.0b013e3182142fd2](https://doi.org/10.1097/MPA.0b013e3182142fd2)]

37. Masaki Y, Sugai S, Umehara H. IgG4-related diseases including Mikulicz's disease and sclerosing pancreatitis: diagnostic insights. *Journal of Rheumatology* 2010;**37**(7):1380-5. [DOI: [10.3899/jrheum.091153](https://doi.org/10.3899/jrheum.091153)]

38. Kawano M, Saeki T, Nakashima H, Nishi S, Yamaguchi Y, Hisano S, et al. Proposal for diagnostic criteria for IgG4-related kidney disease. *Clinical & Experimental Nephrology* 2011;**15**(5):615-26. [DOI: [10.1007/s10157-011-0521-2](https://doi.org/10.1007/s10157-011-0521-2)]

39. Ohara H, Okazaki K, Tsubouchi H, Inui K, Kawa S, Kamisawa T, et al. Clinical diagnostic criteria of IgG4-related sclerosing cholangitis 2012. *Journal of Hepato-Biliary-Pancreatic Sciences* 2012;**19**(5):536-42. [DOI: [10.1007/s00534-012-0521-y](https://doi.org/10.1007/s00534-012-0521-y)]

40. Goto H, Takahira M, Azumi A. Diagnostic criteria for IgG4-related ophthalmic disease [Erratum in: *Jpn J Ophthalmol*. 2015 May;**59**(3):201. doi: 10.1007/s10384-015-0376-2]. *Japanese Journal of Ophthalmology* 2015;**59**(1):1-7. [DOI: [10.1007/s10384-014-0352-2](https://doi.org/10.1007/s10384-014-0352-2)]

41. Matsui S, Yamamoto H, Minamoto S, Waseda Y, Mishima M, Kubo K. Proposed diagnostic criteria for IgG4-related respiratory disease. *Respiratory Investigation* 2016;**54**(2):130-2. [DOI: [10.1016/j.resinv.2015.09.002](https://doi.org/10.1016/j.resinv.2015.09.002)]

42. Mizushima I, Kasashima S, Fujinaga Y, Notohara K, Saeki T, Zen Y, et al. Clinical and pathological characteristics of IgG4-related periaortitis/periarteritis and retroperitoneal fibrosis diagnosed based on experts' diagnosis. *Annals of Vascular Diseases* 2019;**12**(4):460-72. [DOI: [10.3400/avd.19-00085](https://doi.org/10.3400/avd.19-00085)]

43. van Vollenhoven RF, Mosca M, Bertsias G, Isenberg D, Kuhn A, Lerström K, et al. Treat-to-target in systemic lupus erythematosus: recommendations from an international task force. *Annals of the Rheumatic Diseases* 2014;**73**(6):958-67. [DOI: [10.1136/annrheumdis-2013-205139](https://doi.org/10.1136/annrheumdis-2013-205139)]

44. Merkel PA, Herlyn K, Mahr AD, Neogi T, Seo P, Walsh M, et al. Progress towards a core set of outcome measures in small-vessel vasculitis. Report from OMERACT 9. *Journal of Rheumatology* 2009;**36**(10):2362-8. [DOI: [10.3899/jrheum.090373](https://doi.org/10.3899/jrheum.090373)]

45. Merkel PA, Aydin SZ, Boers M, Direskeneli H, Herlyn K, Seo P, et al. The OMERACT core set of outcome measures for use in clinical trials of ANCA-associated vasculitis. *Journal of Rheumatology* 2011;**38**(7):1480-6. [DOI: [10.3899/jrheum.110276](https://doi.org/10.3899/jrheum.110276)]

46. Carruthers MN, Stone JH, Deshpande V, Khosroshahi A. Development of an IgG4-RD responder index. *International Journal of Rheumatology* 2012;**2012**:259408. [DOI: [10.1155/2012/259408](https://doi.org/10.1155/2012/259408)]
47. Wallace ZS, Khosroshahi A, Carruthers MD, Perugino CA, Choi H, Campochiaro C, et al. An international multispecialty validation study of the IgG4-related disease responder index. *Arthritis Care & Research* 2018;**70**(11):1671-8. [DOI: [10.1002/acr.23543](https://doi.org/10.1002/acr.23543)]
48. US Food and Drug Administration. Guidance for industry on patient-reported outcome measures: use in medical product development to support labeling claims; availability. 2009. Available from <https://www.federalregister.gov/documents/2009/12/09/E9-29273/guidance-for-industry-on-patient-reported-outcome-measures-use-in-medical-product-development-to>.
49. Ware JE Jr, Sherbourne CD. The MOS 36-item short-form health survey (SF-36): I. Conceptual framework and item selection. *Medical Care* 1992;**30**(6):473-83. [PMID: 1593914]
50. Herdman M, Gudex C, Lloyd A, Janssen MF, Kind P, Parkin D, et al. Development and preliminary testing of the new five-level version of EQ-5D (EQ-5D-5L). *Quality of Life Research* 2011;**20**(10):1727-36. [DOI: [10.1007/s11136-011-9903-x](https://doi.org/10.1007/s11136-011-9903-x)]
51. ICH 1994. ICH harmonised tripartite guideline. Clinical safety data management: definitions and standard for expedited reporting E2A. Step 4 version. In: International Conference on Harmonisation of Technical Requirements for Registration of Pharmaceuticals for Human Use. 1994. Available from https://database.ich.org/sites/default/files/E2A_Guideline.pdf.
52. Covidence. Version accessed 4 August 2025. Melbourne, Australia: Veritas Health Innovation, 2025. Available at <https://covidence.org>.
53. Parmar MK, Torri V, Stewart L. Extracting summary statistics to perform meta-analyses of the published literature for survival endpoints. *Statistics in Medicine* 1998;**17**(24):2815-34. [DOI: [10.1002/\(sici\)1097-0258\(19981230\)17:24%3C2815::aid-sim110%3E3.0.co;2-8](https://doi.org/10.1002/(sici)1097-0258(19981230)17:24%3C2815::aid-sim110%3E3.0.co;2-8)]
54. Review Manager (RevMan). Version 9.7.1. The Cochrane Collaboration, 2025. Available at <https://revman.cochrane.org>.
55. Sterne JA, Savović J, Page MJ, Elbers RG, Blencowe NS, Boutron I, et al. RoB 2: a revised tool for assessing risk of bias in randomised trials. *BMJ* 2019;**366**:l4898. [DOI: [10.1136/bmj.l4898](https://doi.org/10.1136/bmj.l4898)]
56. Higgins JP, Savović J, Page MJ, Elbers RG, Sterne JA. Chapter 8: Assessing risk of bias in a randomized trial (chapter last updated October 2019). In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.5 (updated August 2024). Cochrane, 2024. Available from <https://www.training.cochrane.org/handbook>.
57. Schünemann HJ, Vist GE, Higgins JP, Santesso N, Deeks JJ, Glasziou P, et al. Chapter 15: Interpreting results and drawing conclusions (last updated August 2023). In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.5 (updated August 2024). Cochrane, 2024. Available from <https://www.training.cochrane.org/handbook>.
58. GRADEpro GDT. Version accessed 5 August 2025. Hamilton (ON): McMaster University (developed by Evidence Prime), 2025. Available at <https://grade.pro.org>.
59. Tierney JF, Stewart LA, Ghersi D, Burdett S, Sydes MR. Practical methods for incorporating summary time-to-event data into meta-analysis. *Trials [Electronic Resource]* 2007;**7**(8):16. [DOI: [10.1186/1745-6215-8-16](https://doi.org/10.1186/1745-6215-8-16)]
60. Colangelo KJ, Pope JE, Peschke C. The minimally important difference for patient reported outcomes in systemic lupus erythematosus including the HAQ-DI, pain, fatigue, and SF-36. *Journal of Rheumatology* 2009;**36**(10):2231-7. [DOI: [10.3899/jrheum.090193](https://doi.org/10.3899/jrheum.090193)]
61. Page MJ, Higgins JP, Sterne JA. Chapter 13: Assessing risk of bias due to missing evidence in a meta-analysis (chapter last updated August 2024). In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.5 (updated August 2024). Cochrane, 2024. Available from <https://www.training.cochrane.org/handbook>.
62. Campbell M, McKenzie JE, Sowden A, Katikireddi SV, Brennan SE, Ellis S, et al. Synthesis without meta-analysis (SWiM) in systematic reviews: reporting guideline. *BMJ* 2020;**368**:l6890. [DOI: [10.1136/bmj.l6890](https://doi.org/10.1136/bmj.l6890)]
63. Deeks JJ, Higgins JP, Altman DG, McKenzie JE, Veroniki AA, editor(s). Chapter 10: Analysing data and undertaking meta-analyses (chapter last updated November 2024). In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.5 (updated August 2024). Cochrane, 2024. Available from <https://www.training.cochrane.org/handbook>.
64. Schünemann H, Brożek J, Guyatt G, Oxman A, editor(s). Handbook for grading the quality of evidence and the strength of recommendations using the GRADE approach (updated October 2013). GRADE Working Group, 2013. Available from <https://gdt.grade.pro.org/app/handbook/handbook.html>.
65. Schünemann HJ, Higgins JP, Vist GE, Glasziou P, Akl EA, Skoetz N, et al. Chapter 14: Completing 'Summary of findings' tables and grading the certainty of the evidence (chapter last updated August 2023). In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.5 (updated August 2024). Cochrane, 2024. Available from <https://www.training.cochrane.org/handbook>.